

GEPPU: A Self-Paced Reading and Eye-Tracking Benchmark Dataset of German Sentence Processing

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Workshop on Cross-Linguistic and Cross-Dialectal Differences in Language Processing

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Why a controlled German benchmark?

Benchmarks for sentence processing

- ▶ Theories of sentence processing are tested against **one construction at a time**, in studies with their own materials, task and sample (typically a few dozen participants)

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Only a **controlled factorial manipulation** isolates the contribution of one linguistic factor while holding the others fixed.

- ▶ Large-scale experimental datasets are rare, and only one spans a **battery of designs**: the **SAP Benchmark** (Huang et al. 2024; Timkey et al. 2026)
 - self-paced reading from 2,000 participants, eye tracking from 368 participants
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Controlled benchmarks: the state of the art

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- ▶ Beyond that benchmark there is nothing comparable:
four families of phenomena, in one language
- ▶ Phenomena that shaped processing theory just as much have **no benchmark at all**, among them memory-based **interference** and **local coherence**

The cross-linguistic (and cross-dialectal?) gap

- ▶ The controlled literature on other languages is large, but it is **not aggregated**
- ▶ Hundreds of separate studies, each with its own materials, task, sample size and analysis
- ▶ Estimates obtained that way cannot be pooled into one evaluation set

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- ▶ Theories of sentence processing are **not theories of English**
- ▶ With benchmark-scale evidence for one language only, a general mechanism cannot be told apart from a property of that language
- ▶ Case marking and relatively free word order let **German** carry designs that English cannot

The GEPPU benchmark

Ten experimental designs, one participant sample

Code	Phenomenon	Factor A × Factor B	(Roughly) Follows ...
Eight 2 × 2 designs			
GP-SDO	Garden path: subject vs. direct object	Ambiguous/Unambiguous × S-O/O-S	Meng & Bader (2000a)
GP-SIO	Garden path: subject vs. indirect object	Ambiguous/Unambiguous × Active/Passive	Meng & Bader (2000b)
GP-MODIO	Garden path: modifier vs. indirect object	Modifier/No-modifier × Ambiguous/Unambig.	van Kampen (2001)
GP-&	Garden path: coordination ambiguity	NP-/VP-coordination × AP-/PP-modifier	Konieczny et al. (2000)
AGR	Agreement attraction (grammatical)	Sg./Pl. controller × Match/Mismatch	Häussler (2009)
LC	Local coherence	Coherent/Incoherent × Intervener/None	Paape & Vasishth (2016)
SBI	Similarity-based interference	Subject cue × Animacy cue	Schoknecht et al. (2025)
SO	Subject vs. object relative clauses	Subject/Object × Double/Single embed.	Schriefers et al. (1995)
Two 3 × 1 designs			
AA-SYN	Attachment ambiguity, syntactic	High / Low / Ambiguous attachment	van Gompel et al. (2000)
AA-SEM	Attachment ambiguity, semantic	High / Low / Ambiguous attachment	Traxler et al. (1998)

All ten designs were run on the **same participants under one procedure**, so the phenomena can be compared directly, without the usual between-study confounds.

Every contrast had a pre-registered direction

	A	B	A×B
GP-SDO	+	—	—
GP-SIO	+	—	—
GP-MODIO	—	+	—
GP-&	—	—	+
AGR	—	+	0
LC	+	0	—
SBI	+	+	0
SO	—	+	—
	Amb?	High?	
AA-SYN	—	0	
AA-SEM	—	0	

- ▶ Derived from the literature and **pre-registered before data collection**, with the full analysis plan (osf.io/wpra9)
- ▶ Where theories conflict, we **committed to one direction** anyway (AGR: cue-based retrieval vs. feature percolation)
- ▶ One **target contrast** per design, **shaded** in the table
- ▶ All of our literature-derived predictions are **qualitative**, so they fix a direction but leave the magnitude open
- ▶ Every results figure also carries a second, quantitative prediction, from **German GPT-2 surprisal** computed on the same contrasts

The materials

GP-SDO: subject vs. direct-object ambiguity

Case-ambiguous initial NP, disambiguated only at the verb Meng & Bader (2000a)

[+amb+S0] subject-object order, the preferred resolution

Alle wollten wissen,	welche Lokalpolitikerin	die amtierenden Minister	gelobt	hat,	als	die Pressekonferenz	begann.
all wanted to.know,	which.F.NOM/.ACC local.politician	the incumbent ministers.NOM/.ACC	praised	has,	when	the press.conference	began.

"Everyone wanted to know which local politician praised the incumbent ministers when the press conference began."

[+amb-S0] object-subject order, the dispreferred one

Alle wollten wissen,	welche Lokalpolitikerin	die amtierenden Minister	gelobt	haben,	als	die Pressekonferenz	begann.
all wanted to.know,	which.F.NOM/.ACC local.politician	the incumbent ministers.NOM/.ACC	praised	have,	when	the press.conference	began.

"Everyone wanted to know which local politician the incumbent ministers praised when the press conference began."

Each trial ends with a two-choice comprehension question: *Haben die amtierenden Minister jemanden gelobt?* (Did the incumbent ministers praise somebody?)

AGR: agreement attraction in grammatical sentences

Grammatical sentences with a number-matching or number-mismatching attractor Häussler (2009)

[+sgc+mtch] singular controller, attractor matches in number

Man erinnerte sich daran,	dass	die Tante	des Mädchens	ziemlich lautstark	geklatscht	hat,	als	die Vorführung	stattfand.
one remembered REFL at.it,	that	the aunt	of.the girl	quite loudly	clapped	has,	when	the performance	took.place.

"It was remembered that the aunt of the girl clapped quite loudly when the performance took place."

[+sgc-mtch] singular controller, plural attractor

Man erinnerte sich daran,	dass	die Tante	der Mädchen	ziemlich lautstark	geklatscht	hat,	als	die Vorführung	stattfand.
one remembered REFL at.it,	that	the aunt	of.the girls	quite loudly	clapped	has,	when	the performance	took.place.

"It was remembered that the aunt of the girls clapped quite loudly when the performance took place."

Both sentences are **grammatical**. Cue-based retrieval predicts inhibitory interference under number match.

LC: local coherence

A locally coherent substring inside the sentence, with or without an intervener Paape & Vasishth (2016)

[+lc+intv] locally coherent: *einer der Generäle belagerte* is a well-formed clause on its own

Schließlich wusste man, eventually knew one,	dass that	entlang der Nachschubrouten along the supply.lines	anfänglich initially	einer one	der Generäle of.the generals
belagerte besieged.PPRT/.FIN.SG	Städte cities	verteidigte, defended.FIN.SG,	was which	größere Verluste greater losses	verhinderte. prevented.

"Eventually it was known that initially one of the generals defended besieged cities along the supply lines."

[-lc+intv] the number mismatch blocks the locally coherent parse

Schließlich wusste man, eventually knew one,	dass that	entlang der Nachschubrouten along the supply.lines	anfänglich initially	einige several	der Generäle of.the generals
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"Eventually it was known that initially several of the generals defended besieged cities along the supply lines."

SBI: similarity-based interference

A distractor sharing syntactic or semantic retrieval cues with the target Schoknecht et al. (2025)

[+subj+anim] animate distractor: *der Einbrecher* competes with *der Witwer* at retrieval

Der Nachbar	erfuhr,	dass	der Witwer,	der	erzählt hatte,	dass	
the neighbor	learned,	that	the widower,	who	told had,	that	
der Einbrecher	schrecklich war,	abends regelmäßig	trank,	um	die Geschehnisse	zu vergessen.	
the burglar	awful was,	in.the.evening regularly	drank,	in.order	the events	to forget.	

"The neighbor learned that the widower who had reported that the burglar was awful drank regularly in the evening."

[+subj-anim] inanimate distractor

Der Nachbar	erfuhr,	dass	der Witwer,	der	erzählt hatte,	dass	
the neighbor	learned,	that	the widower,	who	told had,	that	
der Verlust	schrecklich war,	abends regelmäßig	trank,	um	die Geschehnisse	zu vergessen.	
the loss	awful was,	in.the.evening regularly	drank,	in.order	the events	to forget.	

"The neighbor learned that the widower who had reported that the loss was awful drank regularly in the evening."

SO: subject vs. object relative clauses

Subject vs. object relative clause, singly or doubly embedded Schriefers et al. (1995)

[+srel-dbl] subject relative, single embedding

Der junge Beamte,	der	den erfolgreichen Unternehmer,	einlud,	hatte	böse	Absichten.
the young official,	who.NOM	the.ACC successful entrepreneur,	invited,	had	evil	intentions.

"The young official who invited the successful entrepreneur had evil intentions."

[-srel-dbl] object relative, single embedding

Der junge Beamte,	den	der erfolgreiche Unternehmer,	einlud,	hatte	böse	Absichten.
the young official,	who.ACC	the.NOM successful entrepreneur,	invited,	had	evil	intentions.

"The young official who the successful entrepreneur invited had evil intentions."

Both conditions keep the same word order. Only the **two determiners** change.

AA-SYN: attachment ambiguity resolved by syntax

Relative clause attachment forced high or low by gender agreement, or globally ambiguous van Gompel et al. (2000)

[high_syn] the relative clause must attach to the first noun

Der Manager	der Sängerin,	der lange schwieg,	beendete	plötzlich	das Telefonat.
the manager.M	of.the singer.F,	who.M for.a.long.time was.silent,	ended	suddenly	the phone.call.

[low_syn] it must attach to the second noun

Die Managerin	des Sängers,	der lange schwieg,	beendete	plötzlich	das Telefonat.
the manager.F	of.the singer.M,	who.M for.a.long.time was.silent,	ended	suddenly	the phone.call.

[amb_syn] both attachments stay available

Der Manager	des Sängers,	der lange schwieg,	beendete	plötzlich	das Telefonat.
the manager.M	of.the singer.M,	who.M for.a.long.time was.silent,	ended	suddenly	the phone.call.

We predicted an **ambiguity advantage**, with no difference between high and low attachment.

Data collection

Two experiments, the same 114 sentences

Self-paced reading

- ▶ Moving window, one line, segment by segment
(PCIBex; Zehr & Schwarz 2018)
- ▶ Online via **Prolific**, native speakers in Germany or Austria
- ▶ **N = 1,100** tested, **1,024** after exclusions
- ▶ Comprehension accuracy **76.1 %**
- ▶ 30.9 years on average (SD 9.0)

Eye tracking (in progress)

- ▶ **In-lab** at Potsdam, same materials
- ▶ **N = 250** tested of a target 312; 3 excluded
- ▶ Comprehension accuracy **82.4 %**
- ▶ Fixation durations, skip and regression rates are in the usual range for German reading
- ▶ Analyses so far cover **regression path duration** (go-past time)
- ▶ 23.6 years on average (SD 4.7)

Same items, same lists, same questions in both experiments:
The two methods are **directly comparable**, except for their different participant pools.

Design, counterbalancing, exclusions

- ▶ **12 items** per 2×2 design, **9 items** per 3×1 design; every participant sees **three items per condition**
- ▶ Latin square: four lists (1 to 4) for the 2×2 designs crossed with three (x, y, z) for the 3×1 designs, giving **twelve participant groups** G1x...G4z
- ▶ **114 trials** in total, all ten designs **randomly intermixed**, so the designs serve as each other's fillers (no separate filler items)
- ▶ Six practice trials, a break every 38 trials
- ▶ A **two-choice comprehension question after every trial**, targeting the key syntactic dependency

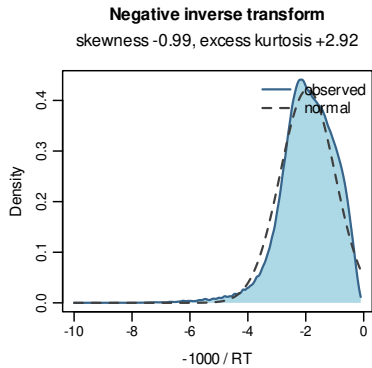
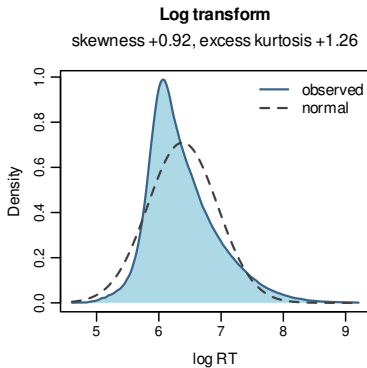
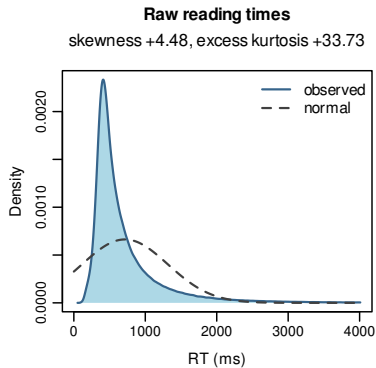
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Exclusions, identical in both experiments: participants below **64 of 108** unambiguous comprehension questions (the six ambiguous-attachment trials have no unique correct answer); SPR reading times < 100 ms or > 10 s.

Do the data behave? Self-paced reading

- ▶ Comprehension accuracy **76.1 %** before exclusions, **77.8 %** after; **715 ms** per segment
- ▶ Models are **log-normal**, as pre-registered; the **negative inverse** would be as defensible ($\hat{\lambda} = -0.5$)



The baseline effects come out as they should

One unit more of...	Change in reading time
characters in the segment	+20.6 ms (+3.5 %)
mean log word frequency (Zipf)	−39.2 ms (−6.2 %)
position within the sentence	−18.8 ms (−3.1 %)
trial number within the session	−1.2 ms (−0.2 %)

Log-normal mixed models with by-participant, by-design and by-item intercepts,
back-transformed to milliseconds at the mean of each predictor.

How each contrast is tested

- ▶ **Log-normal Bayesian hierarchical models** (brms; Bürkner 2017), fitted per design, with the maximal random-effects structure
- ▶ **Critical region** and **spillover region** are separate dependent variables
- ▶ 2×2 : both factors coded ± 0.5 plus their interaction. 3×1 : Ambiguous? $(-\frac{1}{3}, -\frac{1}{3}, +\frac{2}{3})$ and High attachment? $(+\frac{1}{2}, -\frac{1}{2}, 0)$
- ▶ **Bayes factors** via the Savage–Dickey density ratio (Verdinelli & Wasserman 1995), with a **prior sensitivity analysis** on every target parameter: $Normal(0, \{1, 0.5, 0.1, 0.05, 0.01\})$ on the log-ms scale

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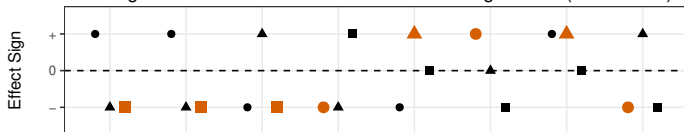
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$BF_{10} > 3$: evidence **for** the effect. $BF_{10} < \frac{1}{3}$: evidence **against** it. In between: undecided.
A design counts as replicating only if the Bayes factor favors the effect **and** the estimate carries the **pre-registered sign**.

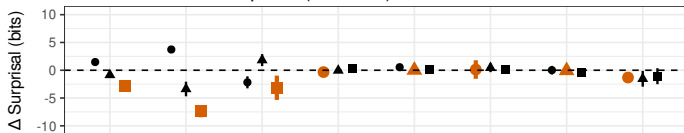
Results

Self-paced reading: predicted vs. observed (critical region)

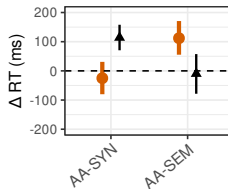
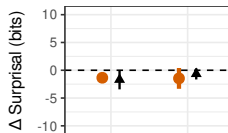
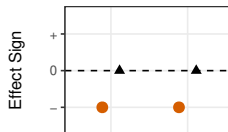
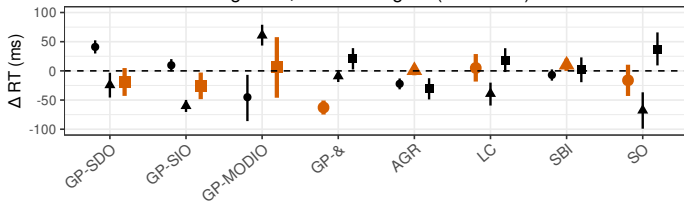
Pre-Registered Predictions From Theoretical Arguments (Qualitative)



Predictions From Surprisal (95% CIs)

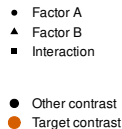
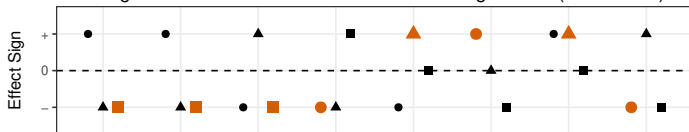


Self-Paced Reading Data, Critical Region (95% Crls)

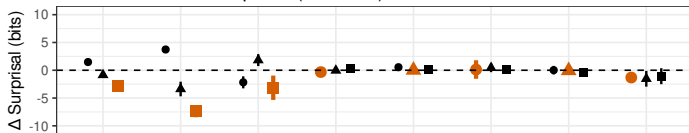


Self-paced reading: spillover region

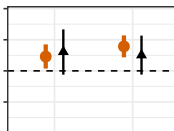
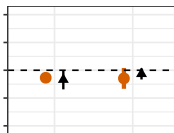
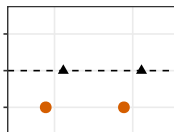
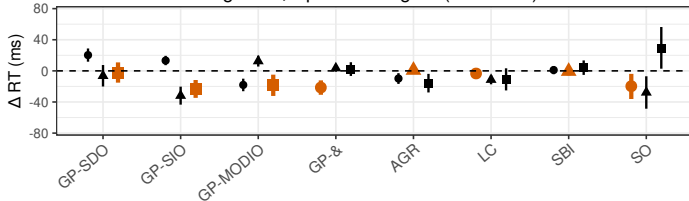
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Predictions From Surprisal (95% CIs)



Self-Paced Reading Data, Spillover Region (95% Crls)



The verdict on ten classic effects (self-paced reading)

Target effect	Design	BF_{10}	Estimate
		critical / spillover	critical / spillover
Effect in both regions	GP-&	$> 100 / 53.3$	-63 ms / -21 ms
Effect in spillover only	GP-SIO	36.9	-23 ms
	GP-MODIO	4.3	-19 ms
Reversed direction	AA-SEM	46.4 / 75.2	+112 ms / +31 ms
Evidence of absence	AGR, GP-SDO, LC, SBI	$< \frac{1}{3}$	≈ 0 ms
Undecided	AA-SYN, SO	0.4 to 2.9	

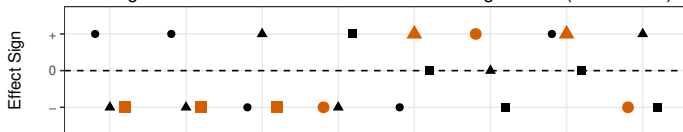
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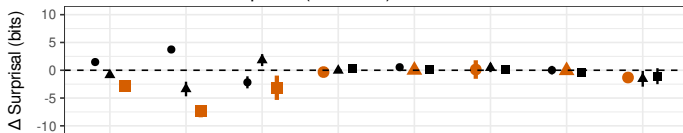
- ▶ No design shows its effect in the critical region alone
- ▶ AA-SEM: *disadvantage*, not the predicted advantage (but note that *disadvantage* has sometimes been found in German when questions target the attachment, as ours do: Logačev & Vasishth, 2016)
- ▶ The confident null results have similar precedents:
 - AGR, GP-SDO weak/absent under comprehension questions (Laurinavichyute & von der Malsburg, 2024; Schoknecht, 2026)
 - LC, SBI show no clear effect on critical verb in German (Paape & Vasishth, 2016; Paape et al., 2021; Schoknecht et al., 2025, 2026)

Eye tracking: regression path duration (critical region; N = 250 of 312)

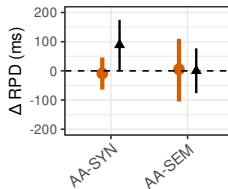
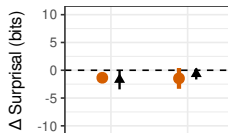
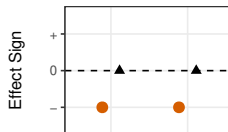
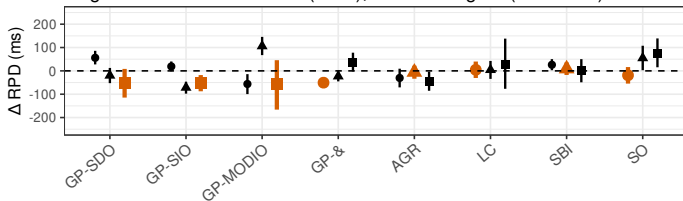
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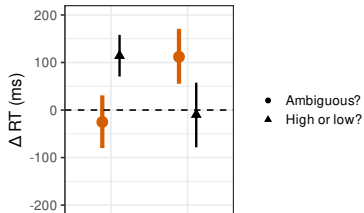
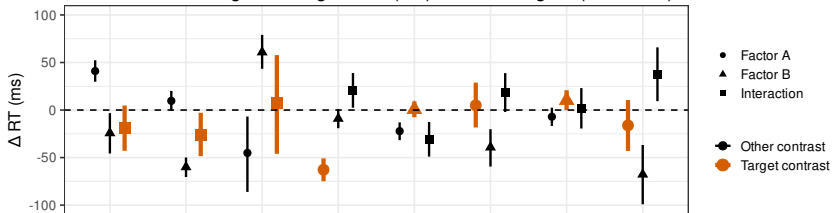


Regression Path Durations (RPD), Critical Region (95% Crls)

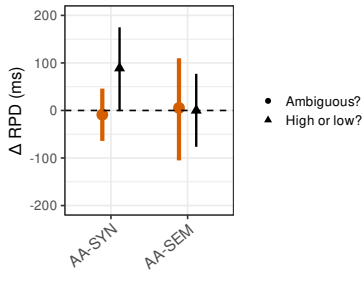
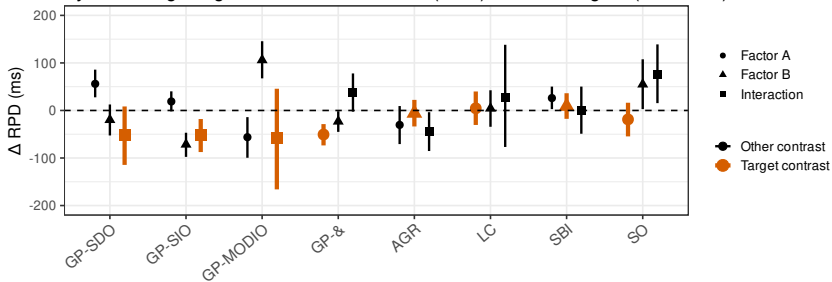


The same designs, the two methods side by side (eye tracking N = 250)

Self-Paced Reading: Reading Times (RT), Critical Region (95% Crls)



Eye Tracking: Regression Path Durations (RPD), Critical Region (95% Crls)



Summary

Summary: What is in GEPPU

- ▶ **Ten classic designs** of German sentence processing, run on the same participants, with the same procedure and the same analysis
- ▶ **Two methods** on the same materials: self-paced reading and eye tracking
- ▶ A **pre-registered directional prediction** for every contrast, fixed before any data were collected
- ▶ Effect estimates and Bayes factors for every main effect and interaction, in two regions
- ▶ Both kinds of predictor, literature-derived and surprisal-derived, can be scored on the same contrasts

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One limit of the design: GEPPU samples **participants broadly** but **items narrowly** (12 items per 2×2 design, 9 items per 3×1 design), so read the nulls in AGR, LC, SBI with that in mind.

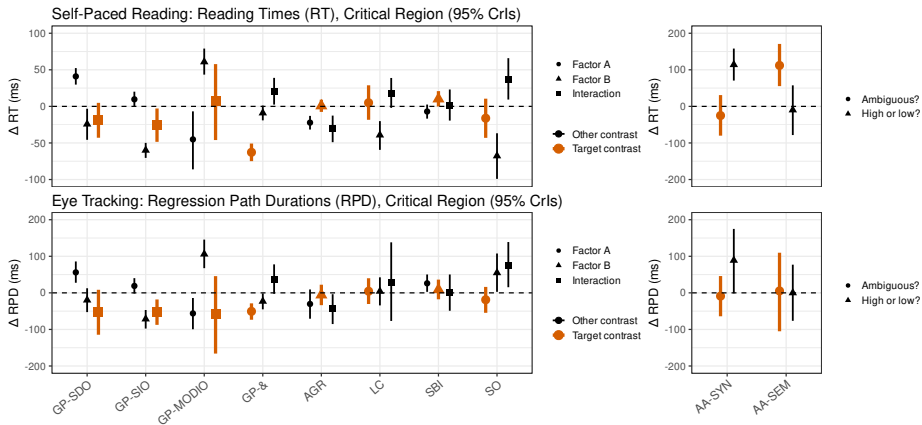
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Pre-registration: osf.io/wpra9. Data and analysis scripts will be released with the paper.

Thank you!



The materials, the data and the analysis scripts will be released with the paper.

Eye tracking is still being collected; the estimates shown are from 250 of a target 312 participants.

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Backup

GP-SIO: subject vs. indirect-object ambiguity

Ambiguous NP read as the subject instead of the indirect object Meng & Bader (2000b)

[+amb+act] active: the ambiguous NP really is the subject

Jeder wusste,	dass	Frau Müller	eine E-Mail	gesendet	hat,	in der	wichtige Ankündigungen	standen.
everybody knew,	that	Ms. <u>NOM</u> /. <u>DAT</u> Müller	an. <u>NOM</u> /. <u>ACC</u> email	sent	has,	in which	important announcements	stood.

"Everybody knew that Ms. Müller sent an email which contained important announcements."

[+amb-act] passive: it turns out to be the indirect object

Jeder wusste,	dass	Frau Müller	eine E-Mail	gesendet	wurde,	in der	wichtige Ankündigungen	standen.
everybody knew,	that	Ms. <u>NOM</u> /. <u>DAT</u> Müller	an. <u>NOM</u> /. <u>ACC</u> email	sent	was,	in which	important announcements	stood.

"Everybody knew that Ms. Müller was sent an email which contained important announcements."

We predicted that the passive costs more **under local ambiguity** (the interaction).

GP-MODIO: modifier vs. indirect-object ambiguity

Genitive (modifier) vs. dative (indirect object) reading of an ambiguous NP van Kampen (2001)

[-poss+amb] no modifier available, so the ambiguous NP must be the indirect object

Dass	die Zofe	der alten Gräfin	den Ring gab,	sah	der Diener	sofort.
that	the maid	to.the/of.the old countess	the ring gave,	saw	the servant	immediately.

"The servant immediately saw that the maid gave the ring to the old countess."

[-poss-amb] unambiguously dative from the start

Dass	die Zofe	dem alten Grafen	den Ring gab,	sah	der Diener	sofort.
that	the maid	to.the old count	the ring gave,	saw	the servant	immediately.

"The servant immediately saw that the maid gave the ring to the old count."

The other two conditions add a possessive modifier (*die Zofe der alten Gräfin dem freundlichen Boten ...*), which makes the genitive reading the correct one.

GP-&: coordination ambiguity

NP–NP vs. elliptical VP–VP coordination, with head position cued by the modifier Konieczny et al. (2000)

[+npco+apmod] NP–NP coordination, the expected resolution (...*and the book to the niece for Christmas*)

Heinz	schenkte	der Nichte	den	Teddy	und	das	
Heinz	gifted	to.the niece	the	teddy.bear	and	the	
spannende und sehr lehrreiche		Buch	zu Weihnachten,	als	er	die Familie	besuchte.
exciting and very instructive		book	for Christmas,	when	he	the family	visited.

[-npco+apmod] VP–VP coordination, the garden path (...*and the book to the nephew*)

Heinz	schenkte	der Nichte	den	Teddy	und	das	
Heinz	gifted	to.the niece	the	teddy.bear	and	the	
spannende und sehr lehrreiche		Buch	dem Neffen,	als	er	die Familie	besuchte.
exciting and very instructive		book	to.the nephew,	when	he	the family	visited.

Disambiguation is **one region before** the critical region: *zu Weihnachten* vs. *dem Neffen*.

AA-SEM: attachment ambiguity resolved by plausibility

As in AA-SYN, but attachment steered by semantic plausibility Traxler et al. (1998)

[high_sem] only the first noun can dance

Die Sängerin	der Ballade,	die bezaubernd tanzte,	wurde	vom Publikum	gemocht.
the singer	of.the ballad,	that charmingly danced,	was	by.the audience	liked.

[low_sem] only the second noun can dance

Die Ballade	der Sängerin,	die bezaubernd tanzte,	wurde	vom Publikum	gemocht.
the ballad	of.the singer,	that charmingly danced,	was	by.the audience	liked.

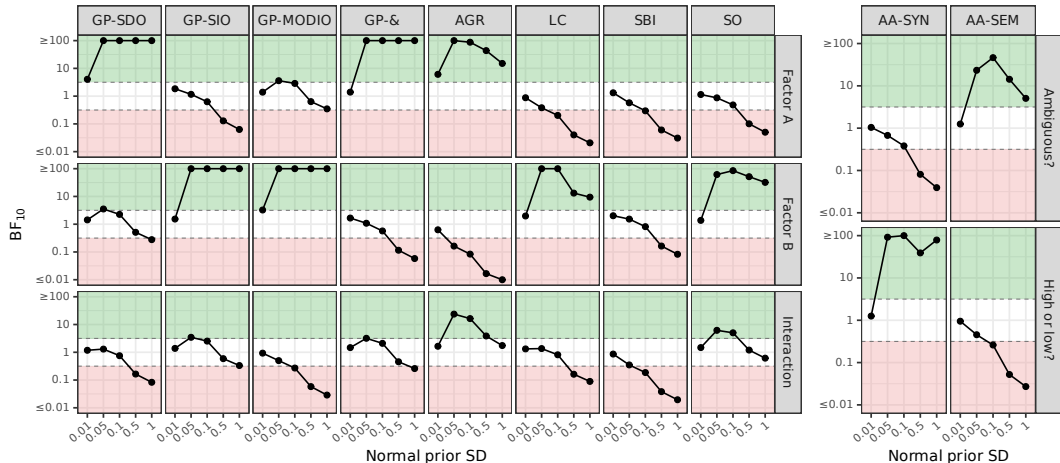
[amb_sem] both nouns are plausible dancers

Die Tochter	der Sängerin,	die bezaubernd tanzte,	wurde	vom Publikum	gemocht.
the daughter	of.the singer,	that charmingly danced,	was	by.the audience	liked.

Here the comprehension question probes the attachment itself: **Wer oder was wurde vom Publikum gemocht?**

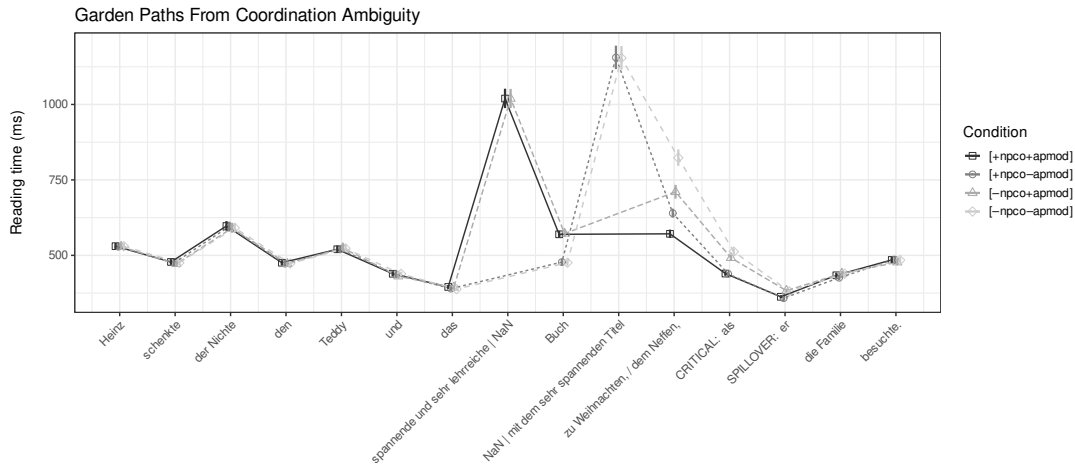
the task condition under which Logačev & Vasishth (2016) found an ambiguity disadvantage in German

How much does the prior matter? (self-paced reading, critical region)



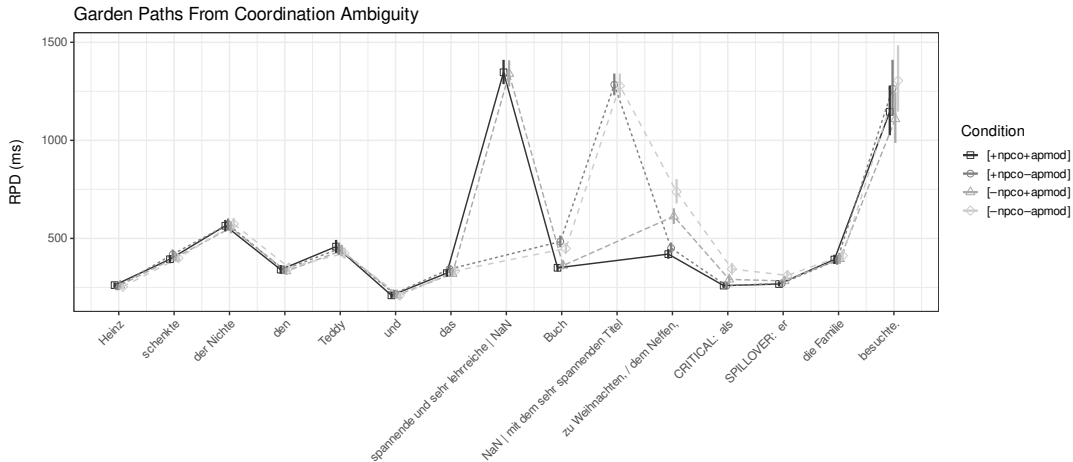
Critical region. **The narrowest prior cannot deliver evidence against an effect**; a conclusion counts only across the four wider ones.

Region by region: GP-& (self-paced reading)



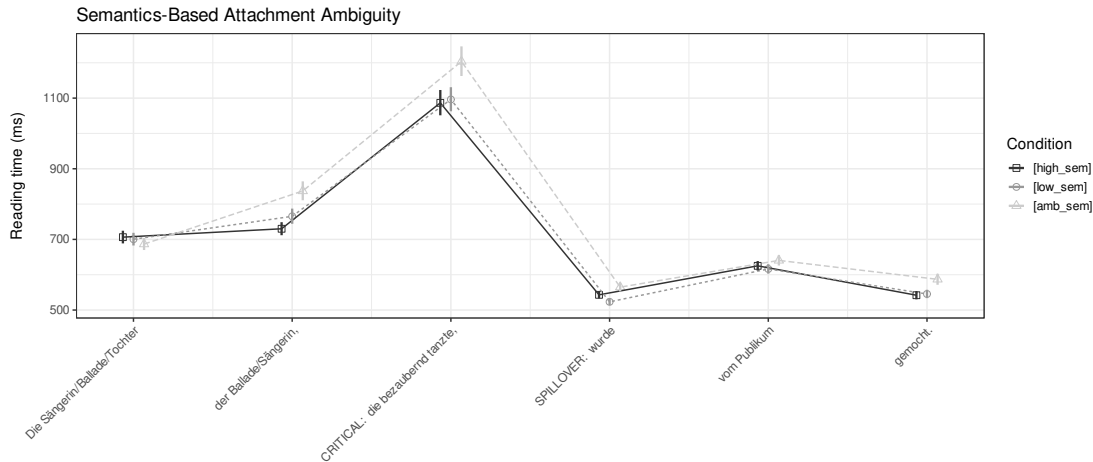
Mean reading time per region and condition, back-transformed from log ms, ± 2 SE.

Region by region: GP-& (eye tracking)



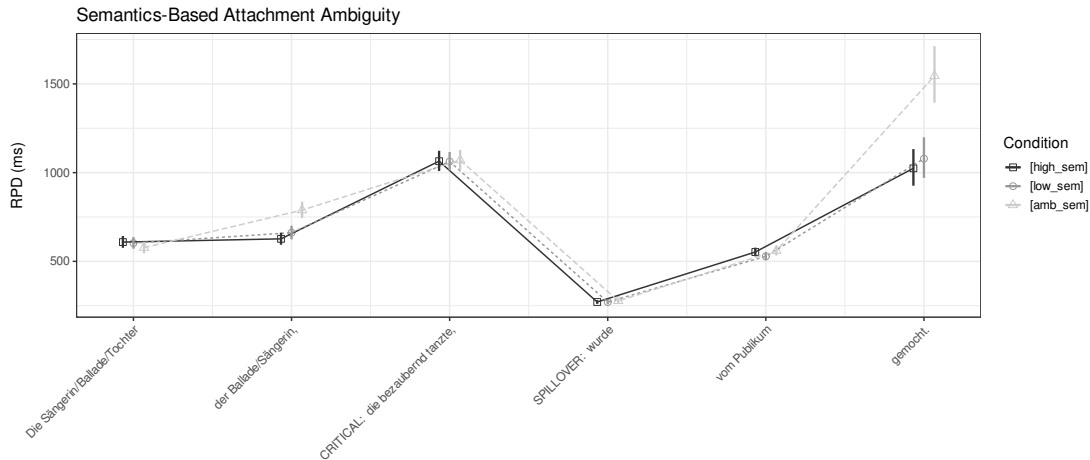
Regression path duration per region and condition. **Preliminary: N = 250.**

Region by region: AA-SEM (self-paced reading)



The design whose effect came out **reversed**.

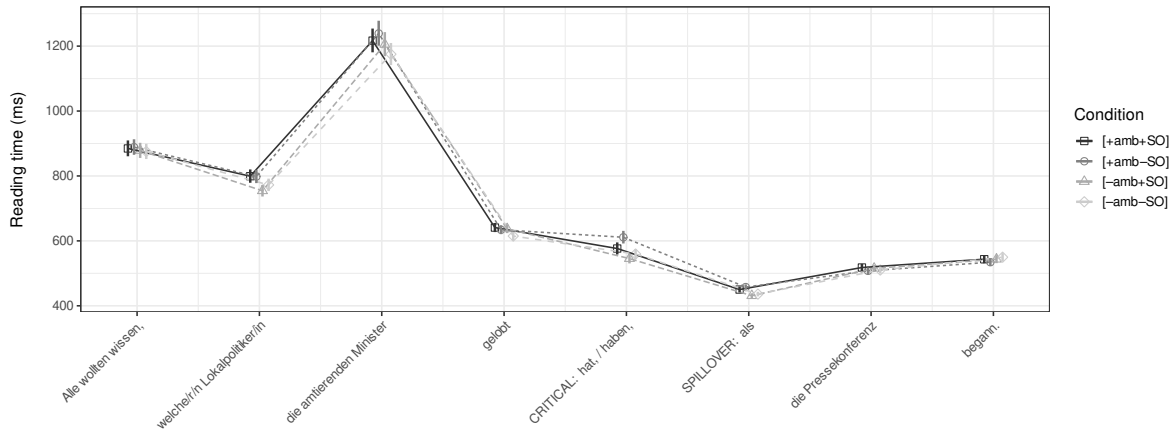
Region by region: AA-SEM (eye tracking)



Preliminary: N = 250.

Region by region: GP-SDO (self-paced reading)

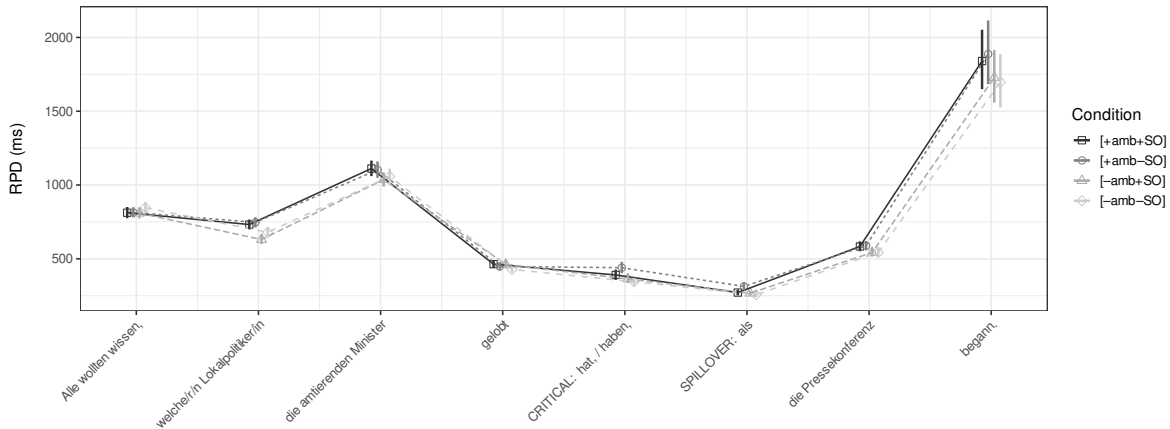
Garden Paths From Subject-vs.-Direct-Object Ambiguity



The design that **failed to replicate** its classic garden path effect.

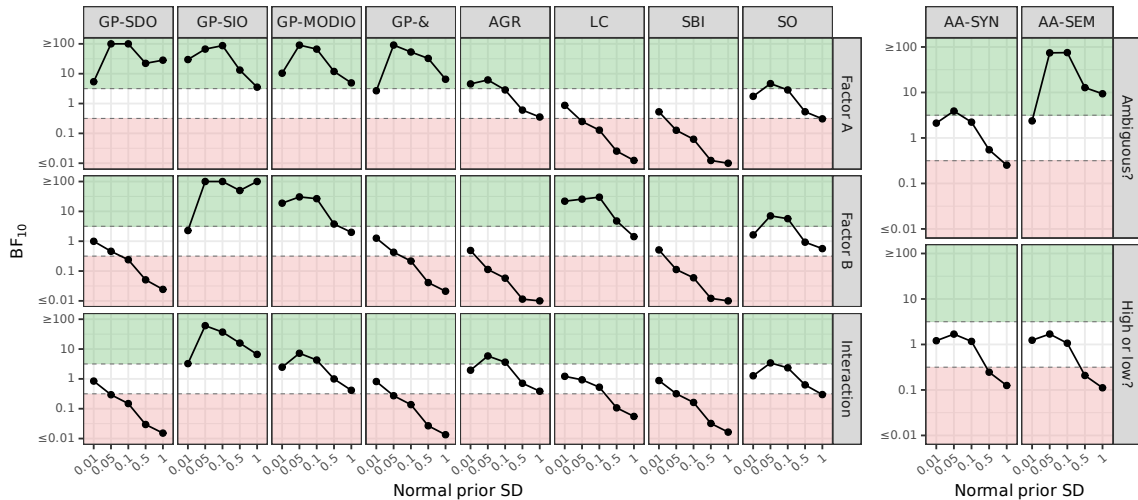
Region by region: GP-SDO (eye tracking)

Garden Paths From Subject-vs.-Direct-Object Ambiguity



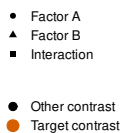
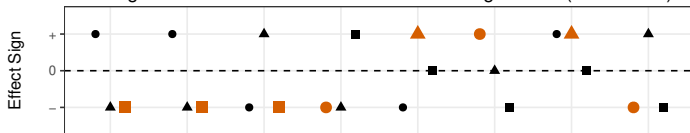
Preliminary: N = 250.

Bayes factors, spillover region (self-paced reading)

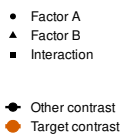
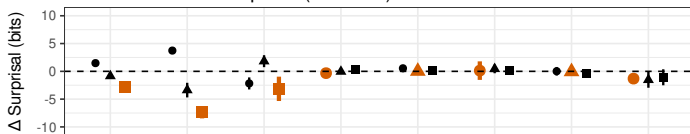


Eye tracking, spillover region (N = 250 of 312)

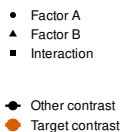
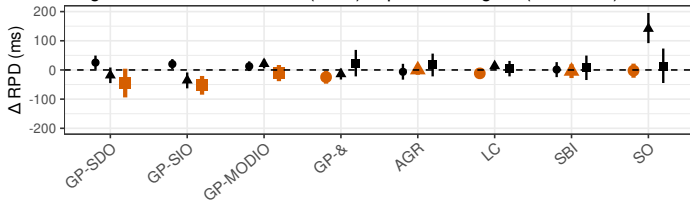
Pre-Registered Predictions From Theoretical Arguments (Qualitative)



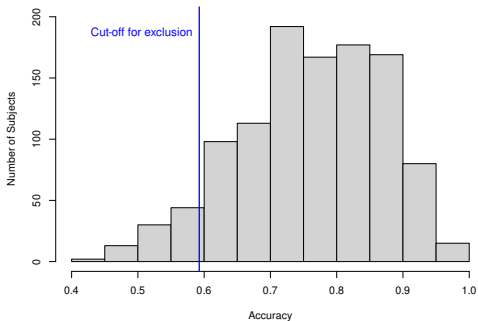
Predictions From Surprisal (95% CIs)



Regression Path Durations (RPD), Spillover Region (95% CIs)



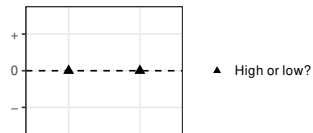
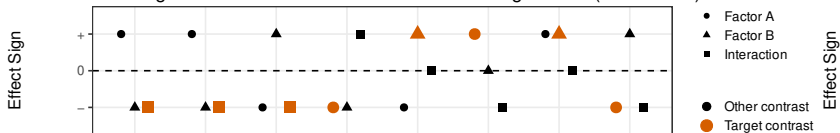
Comprehension accuracy (self-paced reading)



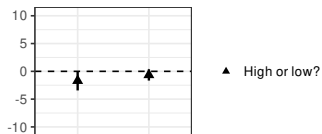
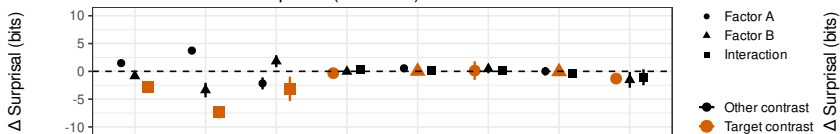
- ▶ Cut-off: 64 correct of 108 unambiguous questions; 76 of 1,100 participants fell below it
- ▶ Lowest per design: GP-SDO and SO at 64.0 %
- ▶ Highest: AGR at 87.2 %, LC at 85.5 %
- ▶ Accuracy effects per design are in the figure on the next frame

Accuracy effects per design (self-paced reading; positive = more errors)

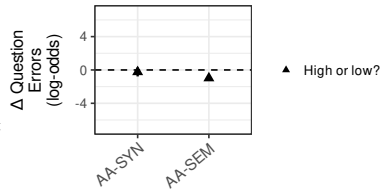
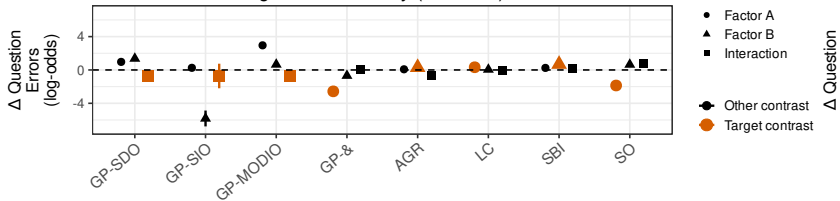
Pre-Registered Predictions From Theoretical Arguments (Qualitative)



Predictions From Surprisal (95% CIs)



Self-Paced Reading Data, Accuracy (95% CIs)



Some effects that we didn't predict (self-paced reading)

- ▶ AGR: the match effect never appeared. The critical region carries an **unpredicted controller number $\times$ match interaction** (BF_{10} up to 23.7)
- ▶ LC: the local coherence effect stayed away. What came out was a **large intervener main effect** (critical region, roughly -40 ms), plausibly a sentence-position effect
- ▶ SO: **doubly embedded relative clauses were read faster** than singly embedded ones (critical region, roughly -68 ms), plausibly a word-position confound
- ▶ SBI: the critical verb shows nothing. An **animacy effect** appears pre-critically, consistent with encoding interference
- ▶ GP-&: disambiguation happens **before** the critical region, so the critical-region estimate understates when the garden path is felt